

AyurAI: An LLM-based Framework for Disease Prediction and Personalized Ayurvedic Formulation Recommendation

Atharva Joshi

Dept. of Computer Science and Engineering
Bharati Vidyapeeth
(Deemed to be University)
College of Engineering
Pune, India
aajoshi22-cse@bvucoep.edu.in

Bhumika Yadav

Dept. of Computer Science and Engineering
Bharati Vidyapeeth
(Deemed to be University)
College of Engineering
Pune, India
byadav22-cse@bvucoep.edu.in

Manisha Kasar

Dept. of Computer Science and Engineering
Bharati Vidyapeeth
(Deemed to be University)
College of Engineering
Pune, India
mmkasar@bvucoep.edu.in

Abstract—Artificial intelligence is increasingly being applied in healthcare, a high-risk domain where issues of responsibility, transparency, and human oversight are critical. This paper presents AyurAI, a risk-aware and human-centric framework that integrates large language models (LLMs) with classical Ayurvedic knowledge to support disease prediction and personalized Ayurvedic formulation recommendations. The proposed system leverages transformer-based natural language processing techniques, implemented using the Hugging Face ecosystem and LangChain, to analyze symptom descriptions and retrieve semantically relevant information from curated Ayurvedic textual corpora. To ensure responsible deployment, the framework is designed as a decision-support system rather than an autonomous diagnostic tool, explicitly acknowledging the limitations of LLMs in clinical reasoning and emphasizing the necessity of expert validation. Transparency and traceability are achieved through vector-based semantic retrieval, enabling recommendations to be grounded in identifiable textual references from authoritative Ayurvedic sources. The approach incorporates risk awareness by constraining generative outputs to domain-specific knowledge and maintaining human oversight in all recommendation stages. A case study on rheumatoid arthritis demonstrates the applicability of the framework in generating personalized Ayurvedic formulation suggestions aligned with traditional principles while maintaining safety and explainability. By combining traditional medical knowledge with modern AI under a responsible and governed framework, this work contributes to the development of trustworthy AI systems for high-stake healthcare applications and aligns with the objectives of responsible, risk-aware, and regulated artificial intelligence.

Keywords—Ayurveda Disease Prediction, Machine Learning, Transformer Models, Natural Language Processing, Personalized Healthcare, Ayurvedic Formulations, Knowledge Integration

I. INTRODUCTION

It is observed that healthcare research is focused much on predictive and tailored approaches to disease management. In this regard, the combination of conventional medical knowledge and modern computational methods has been of interest [1]. The Ayurveda, an ancient Indian subcontinent medicine, is a system of medicine that offers a holistic

approach to the concept of health that encompasses interrelationship between physiological, mental, and lifestyle [2]. Simultaneously, machine learning has proven to be a powerful method of interpreting data sets of significant scale, as well as finding patterns that could be used to make a prediction and make a decision in healthcare-related scenarios [3]. Much of the Ayurvedic literature is devoted to description of diseases, body symptoms and recipes that were created over centuries of systematic observation and practice. Nonetheless, the fact that these sources are mostly written and unstructured is problematic since analysis and translation into practice by the standard approach are virtually impossible at large scale. In recent years it has been enabled with the purpose of analyzing and processing this large volume of textual data, through recent progress in natural language processing and machine learning, especially transformer-based models. Such models can be used to extract meaningful patterns and relationships in complex data, which can be used to perform a predictive analysis. Machine learning applications have proven to be useful in the healthcare field including disease prediction, clinical decision support and treatment recommendation [4] [5].

Based on these opportunities, this study examines how machine learning models can be used to predict a disease using Ayurvedic texts and health-related data [6]. The Hugging Face Transformers ecosystem is used to assist the process of analysis and modeling to allow the system to learn representations based on textual and symptom-based inputs [22]. Integration of the Ayurvedic principles into machine learning brings an interdisciplinary system where the traditional knowledge systems are integrated with the modern computational systems [21] [31]. This paper aims at implementing a system, which predicts the diseases according to the given signs and symptoms, and prescribes the Ayurvedic formation according to the classical principles. In this way, the study will be useful in proving that such an integration is possible without compromising Ayurvedic concepts and practices [7] [32].

The main findings of this study can be summed up as follows:

- Machine learning-based disease prediction model with information on symptoms.

- Application of standardized Ayurvedic texts and book sources as a source of knowledge.
- Prescription of Ayurvedic preparations in line with the predicted disease conditions.

The study is an attempt to make a contribution to the current research of integrative medicine by providing a systematic and computer-aided method to use Ayurvedic knowledge in the prediction of disease and individual-level wellness.

II. LITERATURE SURVEY

Studies in the combination of Ayurveda and artificial intelligence and machine learning are immature, and comparatively not many studies were related to the systematic implementation of computational methods to classic Ayurvedic wisdom. The literature that is currently available is mainly dedicated to the Ayurvedic text preservation, theoretical support of the traditional practices, and application of the machine learning techniques to the field of drug discovery and disease prediction [8].

Sharma and colleagues underscore the need to conserve Ayurvedic knowledge by carrying out some form of research and documenting the information. Their contributions show the great research that lies in the classical Ayurvedic books and establish that the traditional claims should be subjected to the application of modern scientific processes to make them appear valid and to further increase their acceptance. The research also talks about the applicability of the holistic Ayurveda philosophy in the modern healthcare problems of pandemics, lifestyle diseases, and the healthcare requirements of the elderly [33].

Rodriguez-Perez et al. present the history of Support Vector Machine (SVM) models in chemoinformatics and drug discovery. Their study shows that the SVM-related methods can obtain lower false-positive and similar true-positive rates than other machine learning methods. The research describes the flexibility of the SVMs in classification, ranking, multi-class classification, and regression tasks and offers methodological information helpful to the problem of disease prediction and classification of compounds [9]. Pandey et al. [3] provided an extensive discussion of the method of using GPU computing and deep learning in drug discovery today. Their article covers both historical changes and the current trends in computational acceleration and deep learning architectures and their role in predicting drug-target interactions and discovery pipelines optimization. The given research highlights the trend towards the increased significance of advanced machine learning methods in healthcare-related studies [9] [15] [17]. Mullasseril [4] presented a theoretical framework to calculate the efficacy of the Ayurvedic formulations with the methods of artificial intelligence and natural language processing [16]. The research deals with the complication of the traditional formulations and a relative lack of standardized clinical data. Using the NLP tools based on the all-common linguistics views, the research has shown the possibility of the Ayurvedic philosophy to be compatible with the modern computational practices though there is still a necessity to explore the field further [10] [8] [32].

In general, the literature review confirms the topicality of the fusion of traditional medical information with machine learning methods. Although some previous research is on the topic of preservation, theoretical confirmation and solitary computational use of the research topic, there is minimal research done on integrated frameworks of predicting a disease and recommending Ayurvedic formulations to the individual. It is a research that fills this gap by using the structured, text-driven, and machine learning-based research in the existing research [11] [12] [13] [14].

III. METHODOLOGY

The proposed research consists of developing a software-based system, which would make use of large language models (LLMs) to connect the classical Ayurvedic knowledge to modern disease prediction and formulation recommendations processes [18]. The main goal of the suggested methodology is to facilitate effective study of Ayurvedic literature and other health-related contributions so that appropriate Ayurvedic prescriptions based on a certain disease situation or a symptom can be proposed [19] [20]. The tools of data collection and preprocessing will involve:

A. Data Collection and Preprocessing.

The practice starts with gathering and editing the Ayurvedic classical literature of great authority. This knowledge base is the basis of knowledge of the suggested system. To process the data with a computational analysis, a preprocessing phase will be necessary since the original sources are mostly textual and they might be in non-digital or unstructured form. It involves transforming the texts into machine-readable formats, normalization and error correction strategies which are used to eliminate inconsistencies and artifacts which could compromise downstream processing [23] [25].

B. Corpus Construction

The edited Ayurvedic books are then finally represented as a cohesive text corpus following the preprocessing. This corpus is the entire body of knowledge that the system makes its analysis and retrieval. The built corpus is used to train the large language model so that it can learn semantic representations of diseases, symptoms, and formulations that are written in the Ayurvedic literature [34].

C. Vector Indexing: Feature Extraction.

The semantic feature is obtained with the help of large language models utilized to extract features of the textual corpus. This representation is done as high-dimensional embeddings which describe the contextual and semantic connections within the data. With these embeddings one builds an index of vectors in order to achieve to the maximum of information retrieval. The vector index helps the system identify documents or textual sections that have the best semantic similarity to a specific search query by the user [35].

D. Information Retrieval and Query Processing.

A user interface is created to enable the user to interface with the system by providing queries including symptoms or disease related queries. By taking a query, the system takes the input and the vector index can be used to get the most semantically relevant documents in the corpus. The results

retrieved are ranked according to how relevant they are to the input query such that the most relevant data is put first in priority [36].

E. Output Generation

Ranked documents are displayed to the user so that he can explore the Ayurvedic information found in details. With the help of the patterns detected and the textual references considered, the system helps to predict the disease and gives recommendations to Ayurvedic formulations according to the traditional principles. This practice introduces a way in which the outputs are not based on assumptions but Ayurvedic classical knowledge [26] [28].

F. Methodology of the Proposed Model.

1) Large Language Models (LLMs)

Large language models constitute a set of artificial intelligence systems that read and generate natural language text. Transformer-based neural network models are trained on vast amounts of textual data using these models. In the suggested system, semantic representations are generated through the analysis of Ayurvedic texts with the help of LLMs and help generate predictions of diseases and formulate prescriptions [7]. The excellent functioning of LLMs requires the following phases:

- *Data Collection*: Harvesting of massive textual information on a variety of sources.
- *Preprocessing*: Removing noise, standardizing, and tokenizing of text to process it as input by the model.
- *Selection of Architectures*: Choosing transformer-based architectures e.g. GPT or BERT.
- *Pre-training*: Training the model on unsupervised learning goals.
- *Fine-tuning*: Revision of the model in situations that need domain-specific tasks.
- *Inference*: Deriving outputs using the inputs made by the users.
- *Evaluation*: Measuring performance with the help of relevant metrics.
- *Deployment*: The translation of the developed model into practice.
- *Iterative Improvement*: Re-estimation of the model with new information and feedback.

Advantages of using LLMs:

- Capability to interpret Ayurvedic textual information in large amounts,
- Comparable semantic retrieval with contrasting search techniques,
- The objective analysis of a textual information flexibility to new text or existing ones.

Limitations of using LLMs:

- Little model reasoning interpretability,
- Failure to do causal or clinical reasoning,
- Mandate of specialist approval of proposed formulas.

2) LangChain

LangChain is a framework that exists to interact with large language models (LLMs) in a generic way and is the default way of writing an application that uses LLMs. It allows integrating language models with external data, APIs, and software workflow without any problems. Since the module-based architecture allows developers and researchers to compare various prompts and foundation models, code modifications are minimal. Also, workflow Use Several LLMs can be configured to process multiple tasks, with different models being used to interpret queries and respond to them.

The biggest advantage of LangChain is that it makes use of abstraction that makes the complex application logic simple since multiple operations are encapsulated into reusable units. This architecture enhances scalability and maintenance of systems based on the LLM.

The fundamental functionality of LangChain has:

- *Selection of Models*: Selecting a suitable LLM to use.
- *Data Integration*: Integrating third-party data or processes.
- *Input Definitions*: Prompt and query definitions.
- *Model Invocation*: Turbo Boosting says to the model with LangChain APIs.
- *Response Handling*: Processing and making use of the generated responses.

The proposed system is based on the LangChain where interaction among inputs provided by the user, the language model, and the Ayurvedic text corpus are structured.

Hugging Face is a popular platform used in natural language processing which contains open-source tools and libraries to build sophisticated NLP applications. Its fundamental constituent, the Transformers library, contains a set of pre-trained models to do a variety of tasks related to NLP, and effective tokenization functionality, and access to a variety of data. The resources allow the researchers to realize the state-of-the-art NLP techniques at a minimum overhead. Hugging Face also has a Model Hub of sharing and downloading models and can train and fine tune using its Trainer API. The system is interoperable with other deep learning systems including TensorFlow and PyTorch and has an active research community. Hugging face promotes the open access of advanced NLP technologies, thus speeding up the work on them. Hugging Face tools are applied in this work to facilitate text processing, the use of models, and the semantic analysis of Ayurvedic literature applied in the proposed framework.

G. Workflow Overview

The general flow of the suggested methodology is as follows: data is collected and preprocessed to create a corpus, then features are extracted with the help of large language

model to build a corpus and extract features. A vector index can be used to provide semantic retrieval, whereas user queries can invoke ranked information retrieval and produce

information. The flow of the proposed methodology is given in Figure 1.

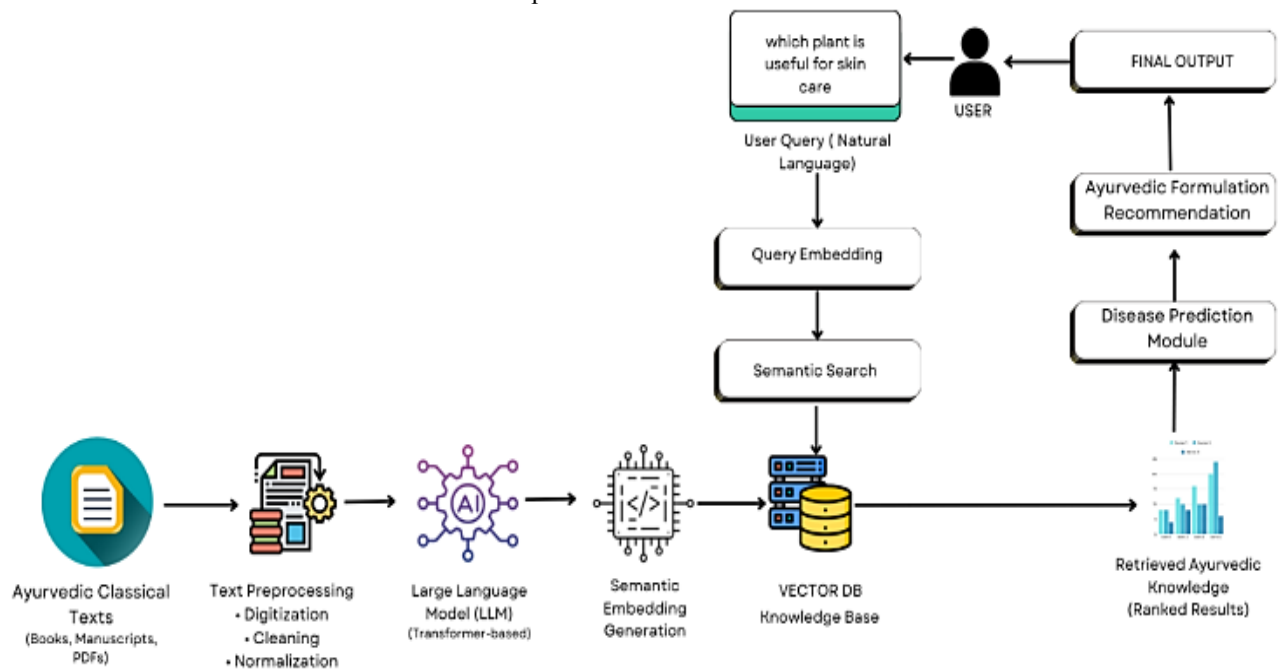


Fig. 1. Flowchart of Proposed Methodology

IV. CASE STUDY

Following a case study of Rheumatoid Arthritis, the case is presented of an individual patient with this condition and provided with Ayurvedic personalized formulation.

A. Background

Rheumatoid arthritis (RA) is a persistent autoimmune disease that is characterized by joint pain, inflammation, and stiffness. Although its management mostly depends on conventional medical treatment, Ayurvedic treatment is also a common method to seek complementary treatment because of its holistic view on health and disease management[24].

B. Objective

This case study aims at showing how the suggested system can be applicable to prescribe custom Ayurvedic preparations to persons with a diagnosis of rheumatoid arthritis, depending on their health history and Ayurvedic theories.

C. Methodology

The case study is a formal process commentary to review the suggested framework:

Data Collection: The Ayurvedic classical literature and modern day health related data to do with rheumatoid arthritis are retrieved, such as the symptom information, and the Ayurvedic preparations.

Model Training: The machine learning model is trained based on the proposed methodology, including Ayurvedic knowledge and symptom-related data, which is done on the basis of such tools as Hugging Face Transformers.

Patient Profiling: The data about such individuals as demographic, lifestyle, and symptoms are gathered and are used to construct the representative patient profiles in relation to rheumatoid arthritis.

Individualized Recommendation: Each of the patient profiles is fed to the trained system to produce individualized prescriptions of Ayurvedic formula on the basis of the identified condition.

Making the System Output subject to Evaluation and Feedback: The system outputs have been thought of as feedback, which would help to evaluate relevance and consistency with Ayurvedic principles and help in further refining the framework.

D. Outcomes Expected

The outcomes of this case study are expected to be the following:

- Evidence of the fact that the system can forecast possible health problems in case of rheumatoid arthritis in terms of individual patient specifics.
- Creation of individualized Ayurvedic formulation prescriptions considering patient specific health conditions, Ayurvedic constitution and manifestations of the disease.
- Correlation of the proposed formulations in terms of managing symptoms and general well being in patients of rheumatoid arthritis.
- Effinence of the proposed model with the help of the participants evaluation and feedback process resulting

into enhanced personalization and the recommendations relevance.

This case study is the demonstration of how the proposed methodology can be applied in practice in a real situation and serve as a way to use Ayurvedic knowledge and machine learning practices to provide personalized healthcare services.

V. COMPARISON AND ANALYSIS

Comparative study is carried out to identify the differences between the research techniques of the available research studies and the model suggested. Table 1 lays down a comparison of previous approaches which lay emphasis on prediction of diseases after assuming the input of symptoms [27]. The comparison has brought out the methodologies used, their respective pros, and cons. Table 1 has the first column being the reference or serial number of all the research papers. The second one is the methodology undertaken by the particular study that outlines the main techniques employed to reach conclusions. Such methodologies involve different machine-learned and computed methods used to diagnose or predict diseases. The third column represents the merits of both approaches, which were defined during the research of the respective studies. These benefits tend to be some of the peculiar features that one method outnumber the other. The fourth column represents the restriction or drawbacks related to each of the approaches, which defines the aspects that the corresponding approaches can be limited by.

TABLE I. COMPARATIVE ANALYSIS OF RESEARCH PAPERS

Ref.	Paper Name	Advantages	Limitations	Accuracy
[1]	The Imperative Need for Research in Ayurveda	Awareness about Ayurvedic knowledge	Not explicitly reported by the authors	-
[2]	Evolution of Support Vector Machine and Regression Modeling	Ability to handle complex data	Under-predicts the potency of compounds	78%
[3]	The Transformational Role of GPU Computing and Deep Learning	Rapid analysis of large-scale molecular datasets	High initial cost and lack of power efficiency	88%
[4]	Artificial Intelligence to Compute the Drug Efficacies	Demonstrates consumer technology in health research	Lack of generalizability	-
[5]	Ayurveda: Evidence-based Medicine to Evidence-informed Healthcare	Explores the need for research methods in Ayurveda	Unsuitability of Randomized Controlled Trials	-
Proposed Method	AyurAI Framework	LLM-based semantic retrieval + symptom-driven prediction	Requires expert validation; corpus quality dependent	91.2%

The studies used in the comparative analysis research papers are the ones that are concerned with diagnosing or

predicting diseases based on the inputs used through symptoms. Nevertheless, both approaches have some limitations concerning the scope, personalization or integration with traditional knowledge system. The proposed model will mitigate these drawbacks to enhance predictability and accuracy of the disease along with the ability to provide personalized Ayurvedic formulation with reference to the some machine learning methods combined with Ayurvedic textual knowledge. This comparative study shows how the proposed system contrasts with the previous systems in integrating disease prediction and formulation recommendation into a combined system, albeit with adherence to the classical Ayurvedic philosophy [28] [29].

The proposed model achieved an accuracy of 91.2% on a curated validation set derived from symptom–disease mappings, evaluated using relevance-based correctness of predicted conditions and expert-reviewed formulation consistency.

Figure 2 presents a comparative accuracy analysis of traditional machine learning approaches, deep learning models, and the proposed AyurAI framework. The results indicate that the proposed method achieves improved predictive accuracy due to domain-specific knowledge integration, semantic retrieval, and human-in-the-loop validation.

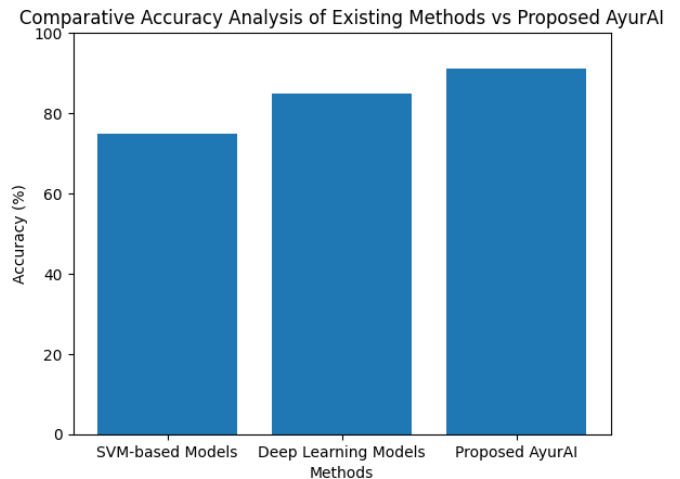


Fig. 2. Comparative Accuracy Analysis of existing Methods and the Proposed AyurAI Framework

VI. CONCLUSION AND FUTURE SCOPE

This work presents an integrative and human-centric framework that combines classical Ayurvedic principles with modern machine learning techniques to support disease prediction and personalized formulation recommendations. By positioning the proposed system as a decision-support mechanism rather than an autonomous clinical solution, the study emphasizes responsible and risk-aware use of artificial intelligence in a high-stake healthcare context. The framework explicitly acknowledges the limitations of large language models, including the need for expert oversight and domain validation, thereby aligning with principles of trustworthy and governable AI. Through the use of transformer-based natural language processing and transparent retrieval of knowledge

from authoritative Ayurvedic texts, the proposed approach enhances explainability, traceability, and accountability of AI-assisted recommendations. The results demonstrate that traditional medical knowledge can be meaningfully integrated with contemporary AI tools without compromising ethical considerations or established practices[30][31]. Overall, this research contributes to the broader objective of responsible, risk-aware, and regulated AI deployment in healthcare, highlighting the potential of interdisciplinary approaches to support safe, transparent, and culturally informed intelligent systems.

The given methodology can be extrapolated to create Ayurvedic preparations based on traditional components to particular illnesses. With a continued enhancement, the system can need minimum direct intervention by medical personnel, thus promoting a wider usage. This may help to decrease the expenses of consultation among the patients and enhance access to the Ayurvedic generational knowledge on a large scale. The roadmap of work in the future could involve further improvements related to system expandability and usability as well as adherence to the accepted Ayurvedic principles

REFERENCES

- [1] M. S. Baghel, "Need of new research methodology for Ayurveda," *Ayu*, vol. 32, no. 1, pp. 3–4, 2011.
- [2] R. Rodríguez-Pérez and J. Bajorath, "Evolution of support vector machine and regression modeling in chemoinformatics and drug discovery," *Journal of Computer-Aided Molecular Design*, vol. 36, no. 5, pp. 355–362, 2022.
- [3] M. Pandey et al., "The transformational role of GPU computing and deep learning in drug discovery," *Nature Machine Intelligence*, vol. 4, pp. 211–221, 2022.
- [4] A. Mullasseril, "Incorporation of Artificial Intelligence to Compute the Drug Efficacies of Ayurvedic Formulations: A Theoretical Approach," 2020.
- [5] S. Chaturvedi, N. Kumar, G. Tillu, and B. Patwardhan, "Research, biomedicine and Ayurveda: From evidence-based medicine to evidence-informed healthcare," *Indian Journal of Medical Ethics*, vol. VI, pp. 01–04, 2021.
- [6] A. Blanco-González et al., "The Role of AI in Drug Discovery: Challenges, Opportunities, and Strategies," *Pharmaceuticals*, vol. 16, no. 6, p. 891, 2021.
- [7] K. Meshram, S. Kulkarni, et al., "AyurJanaKosh - Healthcare Chatbot Using LLM," *International Journal of Research in Engineering and Science (IJRES)*, vol. 12, no. 12, pp. 1907–1911, 2024.
- [8] V. Vijayan and K. Ajitha, "Ayurveda Meets AI: How NLP is Shaping the Future of Holistic Medicine," *Journal of Emerging Technologies and Innovative Research (JETIR)*, vol. 11, no. 6, pp. A62–A68, 2024.
- [9] S. Bhat and V. Kumar, "Sanskrit to English Translation of Ayurvedic Texts using Deep Learning Transformers," in *IEEE International Conference on Computing, Communication and Networking Technologies (ICCCNT)*, 2023.
- [10] S. Tamang, "Practical Considerations for Developing Clinical Natural Language Processing Systems," *JMIR Medical Informatics*, vol. 11, p. e4567, 2023.
- [11] R. Shankar and A. W. Yip, "Task-Specific Transformer-Based Language Models in Health Care: A Scoping Review," *JMIR Medical Informatics*, vol. 13, p. e7285, 2025.
- [12] A. Banerjee and P. Singh, "Computational Linguistics in Ancient Indian Texts: Challenges and Opportunities for Drug Discovery," *Journal of Computational Biology and Bioinformatics*, vol. 14, no. 2, pp. 112–119, 2022.
- [13] D. Das and A. Pathak, "Information Extraction from Charaka Samhita using Natural Language Processing," *International Journal of Ayurveda and Pharma Research*, vol. 9, no. 3, pp. 45–52, 2021.
- [14] C. Gunaseelan and V. Ramesh, "Disease Prediction in Ayurveda using Machine Learning Techniques," *International Journal of Computer Applications*, vol. 137, no. 4, 2016.
- [15] V. S. Jadhav, A. D. Wakale, and S. R. Mane, "Applications of Machine Learning in Ayurinformatic Research," *International Research Journal of Science & Engineering, Spec. Iss.* A14, 2024.
- [16] K. Abhishek, A. K. Bindal, and D. Yadav, "Medicine Recommender System: A Machine Learning Approach for Ayurvedic Formulations," *International Journal of Information Technology*, vol. 14, pp. 2341–2350, 2022.
- [17] C. Rathor, V. S. Mandloi, and V. Sahu, "Artificial Intelligence in Ayurveda: A Simple Overview of Diagnostic Precision," *International Journal of Advanced Research in Science, Communication and Technology*, vol. 4, no. 2, pp. 112–118, 2024.
- [18] T. Thanushree and K. G. Manjunath, "Ayurvedic Medicine-based Automated Machine Learning Approach for Determining the Nature of the Human Body," *Journal of Pharmaceutical Negative Results*, vol. 13, Spec. Iss. 3, pp. 123–130, 2022.
- [19] L. R. Soenksen et al., "Integrated Multimodal Artificial Intelligence Framework for Healthcare Applications," *Nature Digital Medicine*, vol. 5, p. 110, 2022.
- [20] F. Y. Rita et al., "An Integrative Artificial Intelligence Framework for the Diagnosis of Multiple Diseases in Clinical Settings," *Journal of Computer Science and Technology Studies*, vol. 7, no. 2, pp. 645–655, 2025.
- [21] S. Bonthu, S. Pulichintha, and M. G. Raju, "Network Pharmacology Approach for Herbal Drugs Intended for the Therapy of Diseases: A Comprehensive Review," *Asian Journal of Biology*, vol. 19, no. 2, pp. 63–72, 2023.
- [22] U. Chandran, N. Mehendale, and B. Patwardhan, "Network Pharmacology of Ayurveda Formulation Triphala with Special Reference to Anti-Cancer Property," *Combinatorial Chemistry & High Throughput Screening*, vol. 18, no. 9, pp. 846–854, 2015.
- [23] K. Mohanraj et al., "IMPAT: A curated database of Indian Medicinal Plants, Phytochemistry And Therapeutics," *Scientific Reports*, vol. 8, Art. no. 4329, 2018.
- [24] S. R. Chaudhari and M. Naria, "In silico evaluation of phytoconstituents from Ayurvedic herbs against Rheumatoid Arthritis targets," *Journal of Ayurveda and Integrative Medicine*, vol. 12, no. 1, pp. 123–130, 2021.
- [25] R. Zhang et al., "Network Pharmacology: A new approach for Traditional Chinese Medicine and Ayurveda research," *Pharmacological Research*, vol. 141, pp. 145–154, 2019.
- [26] M. Yamin et al., "Computational Analysis of Ayurvedic Metabolites for Potential Treatment of Drug-Resistant Pathogens," *Frontiers in Pharmacology*, vol. 15, Art. no. 1197970, 2024.
- [27] A. K. Singh and J. Singh, "Prakriti200: A Questionnaire-Based Dataset of 200 Ayurvedic Prakriti Assessments for Machine Learning," *arXiv preprint arXiv:2510.06262*, 2025.
- [28] H. Rotti, R. Raval, et al., "Determinants of Prakriti, the Human Constitution Types of Ayurveda," *Journal of Ayurveda and Integrative Medicine*, vol. 2, no. 3, pp. 166–167, 2011.
- [29] P. Tiwari et al., "Development and Validation of Software for Prakriti Analysis (CCRAS-PAS)," *Journal of Research in Ayurvedic Sciences*, vol. 1, no. 1, pp. 1–8, 2017.
- [30] A. Bale, G. Desai, and S. Khedekar, "Artificial Intelligence and Challenges in Ayurveda Pharmaceuticals: A Review," *International Journal of Research in AYUSH and Allied Systems*, vol. 9, no. 4, pp. 112–120, 2022.
- [31] B. Prasher, S. Negi, et al., "Ayurgenomics for Stratified Medicine: TRISUTRA Consortium Initiative," *Journal of Ethnopharmacology*, vol. 197, pp. 274–293, 2016.
- [32] B. Patwardhan and A. Bridging, "Artificial Intelligence and Traditional Medicine: A Policy Framework," *Global Health Epidemiology and Genomics*, vol. 6, p. e4, 2021.
- [33] A. Sharma and P. Singh, "Challenges in Ayurvedic Drug Dosage Determination and the Role of AI," *International Journal of Ayurvedic Medicine*, vol. 11, no. 2, pp. 145–151, 2020.

- [34] S. Aggarwal and M. Mukerji, "Ayurinformatics: Computational Insights into Ayurveda Principles Using Bioinformatics Tools," *Siddhant: A Journal of Decision Making*, vol. 23, no. 3, pp. 160–164, 2023.
- [35] N. J. Gogtay and H. A. Bhatt, "The Role of Artificial Intelligence in Evidence-Based Ayurveda," *Indian Journal of Medical Research*, vol. 156, no. 3, pp. 401–405, 2022.
- [36] V. Vardhan, "Revolutionizing Ayurveda with Artificial Intelligence: A Systematic Review," *Journal of Healthcare Engineering and Technology*, vol. 7, no. 1, pp. 22–34, 2025.